

1) Convex lens

2) Power is inversely proportional to focal length.
So, a convex lens of focal length 10cm has a greater power

3) Given:

Object is placed 5cm in front of the lens
The image is formed behind the lens at 10cm

$$\therefore u = -5 \text{ cm}$$

$$v = 10 \text{ cm}$$

Since, the image formed behind the mirror and the image formed at a distance double the object placed, the lens is a Convex lens.

Now,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{-5}$$

$$\frac{1}{f} = \frac{1+2}{10}$$

$$\frac{1}{f} = \frac{3}{10}$$

$$f = \frac{10}{3}$$

$\therefore$ The value of its focal length is $\frac{10}{3} \text{ cm}$

4) A Given:

$$v = 50 \text{ cm}$$

$$h_i = h_o$$

$$m = -1$$

$$m = \frac{v}{u}$$

$$-1 = \frac{50}{u}$$

$$u = -50 \text{ cm}$$

Lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \frac{1}{-50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$$f = 25 \text{ cm}$$

Now,

$$P = \frac{1}{f_m} = \frac{1}{25 \times 10^{-2}}$$

$$1D = 1m^{-1}$$

$$4m^{-1} = 4D$$

The object is placed at a distance of 50 cm from the lens and the power of the lens is 4D